

Research and Applications

Information extraction from clinical notes: are we ready to switch to large language models?

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Abstract

Objectives: To assess the performance, generalizability, and computational efficiency of instruction-tuned Large Language Model Meta AI (LLaMA)-2 and LLaMA-3 models compared to bidirectional encoder representations from transformers (BERT) for clinical information extraction (IE) tasks, specifically named entity recognition (NER) and relation extraction (RE).

Materials and Methods: We developed a comprehensive annotated corpus of 1588 clinical notes from 4 data sources—UT Physicians (UTP) (1342 notes), Transcribed Medical Transcription Sample Reports and Examples (MTSamples) (146), Medical Information Mart for Intensive Care (MIMIC)-III (50), and Informatics for Integrating Biology and the Bedside (i2b2) (50), capturing 4 clinical entities (problems, tests, medications, other treatments) and 16 modifiers (eg, negation, certainty). Large Language Model Meta AI-2 and LLaMA-3 were instruction-tuned for clinical NER and RE, and their performance was benchmarked against BERT.

Results: Large Language Model Meta AI models consistently outperformed BERT across datasets. In data-rich settings (eg, UTP), LLaMA achieved marginal gains (approximately 1% improvement for NER and 1.5%–3.7% for RE). Under limited data conditions (eg, MTSamples, MIMIC-III) and on the unseen i2b2 dataset, LLaMA-3-70B improved F1 scores by over 7% for NER and 4% for RE. However, performance gains came with increased computational costs, with LLaMA models requiring more memory and Graphics Processing Unit (GPU) hours and running up to 28 times slower than BERT.

Discussion: While LLaMA models offer enhanced performance, their higher computational demands and slower throughput highlight the need to balance performance with practical resource constraints. Application-specific considerations are essential when choosing between LLMs and BERT for clinical IE.

Conclusion: Instruction-tuned LLaMA models show promise for clinical NER and RE tasks. However, the tradeoff between improved performance and increased computational cost must be carefully evaluated. We release our Kiwi package (<https://kiwi.clinicalnlp.org/>) to facilitate the application of both LLaMA and BERT models in clinical IE applications.

Key words: large language model; information extraction; instruction-tuning.

Introduction

The digital transformation of health-care records into electronic health records (EHRs) represents one of the most significant advancements in the modern health-care system.¹ This transformation has paved the way for approaches to advance patient care and clinical research.² Clinical notes within EHRs are among the most information-rich sources, capturing diagnoses, treatments, outcomes, and more.³ These narrative texts, written by health-care professionals, capture a detailed picture of patient information, including diagnosis, treatments, patient outcomes, and much more.⁴ Despite their potential, the massive, complex and unstructured nature of clinical notes has largely presented significant challenges for their effective use.⁵

Clinical information extraction (IE) transforms unstructured clinical text into structured data using Natural

Language Processing (NLP) techniques like named entity recognition (NER) and relation extraction (RE).^{6–8} More specifically, NER identifies medical entities (eg, “aspirin” as a drug), while RE captures their relationships (eg, “81 mg” as its dosage). Over the past 20 years, most clinical NLP research has focused on clinical IE. For instance, 8 out of 10 challenges in the i2b2/n2c2 series from 2006 to 2019 have centered on clinical IE tasks.^{9–16} Bidirectional encoder representations from transformers-based models have previously demonstrated state-of-the-art (SOTA) performance in clinical IE tasks.^{17,18} However, language variability across clinicians and institutions hinders generalizability, as models often fail when applied beyond their training domain.^{5,19}

Recently, large language models (LLMs) such as generative pretrained transformers and the Large Language Model Meta AI (LLaMA) series^{20–22} have shown phenomenal capabilities in

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generative NLP tasks, including text summarization, question answering, and machine translation.^{23,24} However, their effectiveness in traditional IE remains mixed. In zero-/few-shot settings, LLMs often underperform compared to fine-tuned BERT models,^{25–27} and even when fine-tuned, their gains are usually limited to datasets with similar text types, raising concerns about generalizability.^{28,29} Additionally, many studies focus on narrow tasks (eg, fewer than 10 relation types) or single institutions,^{30–34} and few compare LLMs directly to BERT or assess cross-institutional generalization.^{35,36} Tool availability is also limited, only Hsu and Roberts' LLM-IE supports downstream use but lacks fine-tuned models.³⁷ To the best of our knowledge, no comprehensive study has compared fine-tuned LLMs and BERT models in terms of performance and generalizability across clinical notes from multiple data sources, entity types, and relation types. Moreover, there is a lack of research examining the tradeoffs between model performance, throughput, and energy consumption, which is particularly crucial given the varying computational resources available in different hospitals and research institutions.

To address these challenges, we conducted a comprehensive study comparing instruction-tuned LLaMA and BERT models on clinical NER and RE tasks. Our study makes several key contributions to the clinical IE field. First, we developed one of the largest and most comprehensive cross-institutional annotated corpora for clinical NER and RE, comprising 1588 clinical notes from multiple institutions. This corpus encompasses 4 primary clinical entities and 16 corresponding modifiers, providing a robust foundation for advancing clinical IE. Using the developed corpus, we systematically evaluated the performance of instruction-tuned LLaMA-3 models, demonstrating their superiority over previous SOTA BERT-based models in clinical NER and RE tasks, particularly in low-resource and cross-institutional settings, with F1 score gains of up to 7% for NER and 4% for RE. Beyond performance metrics, we compared LLM-based and BERT-based clinical IE models across additional dimensions, including speed/throughput, computational resource requirements, and energy efficiency, which reveals some constraints of LLM-based IE models on real-world applications. Based on these findings, we provided detailed recommendations for model selection across diverse clinical NER and RE scenarios, bridging the gap between AI research and practical clinical use. Furthermore, we introduce Kiwi, one of the first publicly available LLM-based clinical IE systems. Kiwi provides both BERT-based and LLaMA-based models, allowing users to select the most appropriate tools based on their specific needs, computational resources, and task requirements, and it is freely accessible at <https://kiwi.clinicalnlp.org/>.

Methods

Tasks

Our study includes 2 clinical IE tasks: NER and RE. Named entity recognition identifies entity types and boundaries in text,³⁸ while RE identifies relationships between entity pairs.³⁹ We instruction-tuned LLaMA-2 and LLaMA-3 models and compared them to fine-tuned BERT. We focused on extracting 4 main entities—medical problems, tests, drugs, and other treatments—along with 16 modifiers (eg, severity, temporal, dosage). In total, there are 21 possible relations between main entities and modifiers, as detailed in Table 1.

Annotation methods and guidelines are available in the [Supplementary Material](#) and GitHub repository.

Datasets

We developed a comprehensive, cross-institutional clinical IE corpus comprising both real-world and synthetic notes. It consists of real-world and synthetic clinical notes collected cross-institutionally. The majority of clinical notes (1342, 85%) were from UT Physicians (UTP). In addition, we included 146 notes from Transcribed Medical Transcription Sample Reports and Examples (MTSamples),⁴⁰ 50 from the Medical Information Mart for Intensive Care (MIMIC-III),⁴¹ and 50 from 2010 Informatics for Integrating Biology and the Bedside (2010 i2b2).⁹ Table 1 presents the composition and statistics of the 4 datasets used in this study for NER and RE. Unlike the original i2b2 annotations, we reannotated those notes using our expanded schema with additional entity types and fine-grained modifiers (eg, severity, certainty, temporal), making direct comparison with previous results inappropriate. Dataset composition and annotation comparisons are detailed in Table 1 and Table S5.

Models

Large Language Model Meta AI

We instruction-tuned LLaMA-2-chat and LLaMA-3-instruct models (7B, 13B, and 70B for LLaMA-2-chat; 8B and 70B for LLaMA-3-instruct) using parameter-efficient fine tuning to enhance computational efficiency.⁴² Specifically, we used 4-bit quantized models with QLoRA (LoRA settings: $r=16$, $\alpha=64$, dropout=0.05) and focused on tuning linear layers to reduce computational load.⁴³ Fine-tuning parameters included a learning rate of $2e-4$, 2 epochs, batch size of 4, and warmup ratio of 0.05. Model weights are available at huggingface.co/meta-llama,⁴⁴ and we used the vLLM package with a temperature of 0 for reproducible output generation.⁴⁵

Bidirectional encoder representations from transformers

We fine-tuned BioClinicalBERT for NER and BiomedBERT for RE tasks.^{18,46} Based on our preliminary experiments, these 2 distinct BERT models demonstrated superior performance for their respective tasks, motivating our decision to use them separately. For the NER model, we employed a learning rate of $5e-5$ over 20 epochs with a batch size of 4. The RE model was trained using a learning rate of $1e-6$ over 20 epochs with a batch size of 64.

Instruction format

Our approach uses a unified format for NER and RE (see Figure 1B) as listed below:

Named entity recognition

Large Language Model Meta AI models are instructed to identify main entities in clinical note sentences, tagging them with `<span class='entity_type'>...</span>`, where "entity_type" defines the entity.

Relation extraction

The main entity in the input text is tagged similarly, and LLaMA models identify related modifier entities using the same span format. Full instructions are given in Table S1.

The HTML format was adopted to enable a fair comparison with traditional sequence labeling approaches, which

Table 1. Numbers of documents, entities, and combinations between main and modifier entities for train and test splits across all datasets.

		Train (finetune)			Test			
		UTP	MTSamples	MIMIC-III	UTP	MTSamples	MIMIC-III	i2b2
Documents		1292	96	25	50	50	25	50
Main entity	Modifier							
Problem		85 457	4059	2238	3446	2200	2379	1454
	Negation	46 218	1478	438	1863	780	575	271
	Temporal	2737	101	156	114	65	181	73
	Severity	1544	114	91	53	38	131	37
	Condition	296	17	42	18	20	74	1
	Certainty	551	62	87	15	20	63	12
	Subject	2575	26	20	132	12	26	0
	Body location	24 205	959	720	956	588	726	468
	Course	2971	220	163	114	109	184	93
Test		50 003	1529	2297	2151	828	2542	1309
	Negation	224	11	9	5	7	11	2
	Temporal	24 027	115	2130	1055	30	3057	474
	Lab value	33 009	893	1606	1433	398	1963	919
	Reference range	2287	0	11	44	0	5	0
Drug		13 648	747	1047	568	466	1021	681
	Negation	321	20	7	6	8	15	3
	Temporal	16 109	46	44	728	14	69	60
	Strength	6590	154	492	275	171	436	268
	Dosage	5352	61	258	231	46	248	99
	Duration	514	27	77	21	11	27	32
	Form	12 028	58	396	536	47	359	144
	Frequency	6440	209	521	262	219	434	320
	Route	7079	106	377	301	92	353	260
Other treatment		8208	584	542	329	388	510	405
	Negation	321	12	16	6	5	19	6
	Temporal	1534	53	56	59	38	98	59

Abbreviations: i2b2, Informatics for Integrating Biology and the Bedside; MIMIC-III, Medical Information Mart for Intensive Care; MTSamples, Transcribed Medical Transcription Sample Reports and Examples; UTP, UT Physicians.

require precise entity positions, and to mitigate LLaMA's known sensitivity to instruction design. Our pilot tests also showed this structure improves extraction consistency and boundary recognition.²⁵ Our approach operates at the sentence level; each sentence is treated as an independent input instance for both model training and inference. This design choice ensures consistent preprocessing and enables a fair comparison across model types (note that annotations were also performed at the sentence level, so cross-sentence relations were not included in the gold-standard data).

Evaluation

Performance and significance test

We evaluate RE and NER performance using precision (P), recall (R), and F1 scores under exact and relaxed match criteria. Exact matches require entity types, boundaries, and relations to align perfectly, while relaxed matches require only entity type and relation matches with overlapping boundaries. To assess statistical significance of improvements over BERT, we employed bootstrap resampling with 1000 replicates at the sentence level (sample size = 500 sentences per resample). For each replicate, we computed performance metrics and applied a paired 1-sided Wilcoxon signed-rank test to the resulting distributions. We conducted a power analysis assuming an α of 0.05, effect size of 0.02 (ie, a 2% improvement over BERT), with variance estimated from our preliminary bootstrap results. This analysis indicated that at least 929 replicates were required to achieve 80% power, so we chose 1000 replicates for our analysis to ensure adequate statistical power.

Cross-institution generalizability

We assess cross-institution generalizability using the i2b2 dataset, which contains clinical notes from Partners Healthcare and Beth Israel Deaconess Medical Center. This dataset provides evidence for linguistic and institutional generalizability, making it suitable for testing model performance across institutions. Therefore, we excluded i2b2 from fine-tuning to use it solely for generalizability testing.

Computing costs

We assess each model's computational costs by measuring GPU hours, memory usage, and energy consumption during fine-tuning and inference. GPU hours are calculated by multiplying the number of GPUs by the total processing time. Memory usage reflects the average GPU memory in GB per GPU. Energy consumption is estimated from average GPU power usage and converted to carbon emissions using a factor of 0.39 kg CO₂/kWh.⁴⁷

Throughput

We assess each model's throughput during fine-tuning and inference. For fine-tuning, we measure the total time to train and calculate seconds per note by dividing the total time by the number of notes in the dataset. For inference, we determine the time per note using the test dataset with a batch size of 100 for both LLaMA and BERT models.

Results

Fine-tuning performance

Figure 2 highlights the exact match performance of LLaMA-2, LLaMA-3, and BERT models, with full metrics in Table S2.

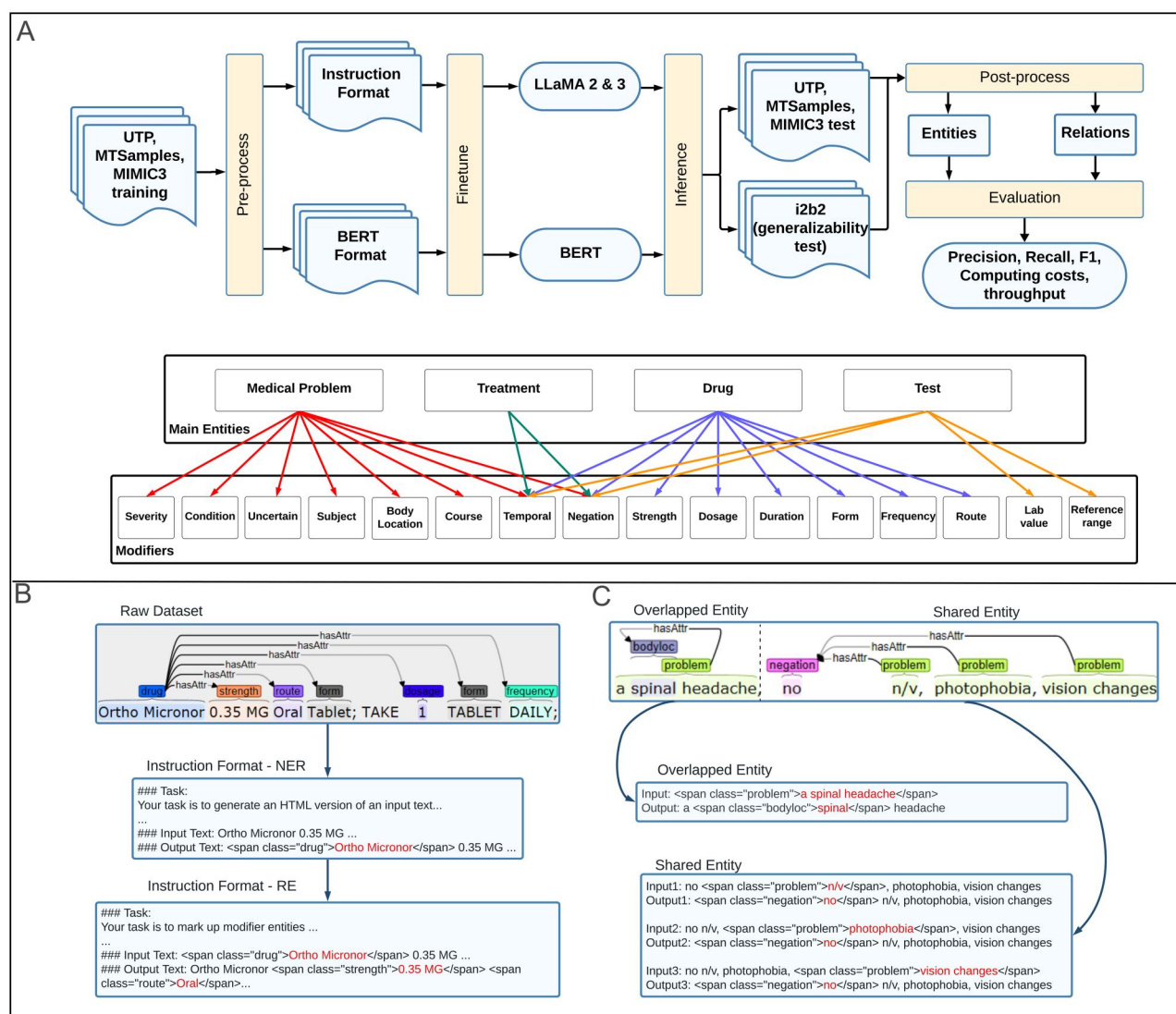


Figure 1. (A) Overview of the study workflow, including entity and relation types in the dataset. (B) Examples of instructions for NER and RE, using a unified format with span tags to highlight all entities and modifiers. (C) The GUI of Kiwi online demo. Abbreviations: i2b2, Informatics for Integrating Biology and the Bedside; MIMIC-III, Medical Information Mart for Intensive Care; MTSamples, Transcribed Medical Transcription Sample Reports and Examples; UTP, UT Physicians.

For all pairwise model comparisons, 95% CIs derived from bootstrapped Wilcoxon tests are provided in [Tables S6 and S7](#). [Figures S1 and S2](#) further provide point plots with 95% CIs for all model-task comparisons. Large Language Model Meta AI models consistently outperformed BERT, particularly with limited fine-tuning data. With sufficient data (eg, UTP dataset), LLaMA models show minor improvements of ~1% on NER and 1.5%-3.7% on RE. However, with less data (eg, MTSamples, MIMIC-III), LLaMA models show greater performance improvement compared to BERT, with 3.7%-4.5% higher F1 scores on NER and up to 12.5% higher on RE.

To explore the performance of LLaMA and BERT in low-resource settings, we conducted dataset-specific fine-tuning by training 2 models—LLaMA-3-70B and BERT—separately on the MTSamples and MIMIC-III datasets. Each model was fine-tuned exclusively on either MTSamples or MIMIC-III, then evaluated across all 4 test sets. Based on the results in [Figure 2](#), where LLaMA-3-70B demonstrated superior performance across datasets, we prioritized using this model for

these experiments. Our findings show that LLaMA-3-70B consistently outperformed BERT across datasets in these low-resource scenarios, as shown in [Tables S3 and S4](#).

Cross-institution generalizability

We evaluated LLaMA models and BERT on cross-institution data using the i2b2 dataset from Partners Healthcare and Beth Israel Deaconess. As shown in [Figure 2](#), LLaMA-3-70B outperformed BERT by over 7% on NER and 4% on RE, demonstrating the superior generalizability of LLaMA models. Please note that the performance of the BERT model on the i2b2 dataset in our study is lower than previously reported results, primarily due to 2 factors: (1) we used an expanded annotation schema with 4 main entity types, increasing task complexity and (2) our model was evaluated in a zero-shot setting, trained without i2b2 dataset. In contrast, prior studies such as Si et al. trained and tested within the i2b2 dataset.¹⁷ When evaluated in-domain (eg, on UTP), our BERT model achieves strong performance (F1 = 0.921).

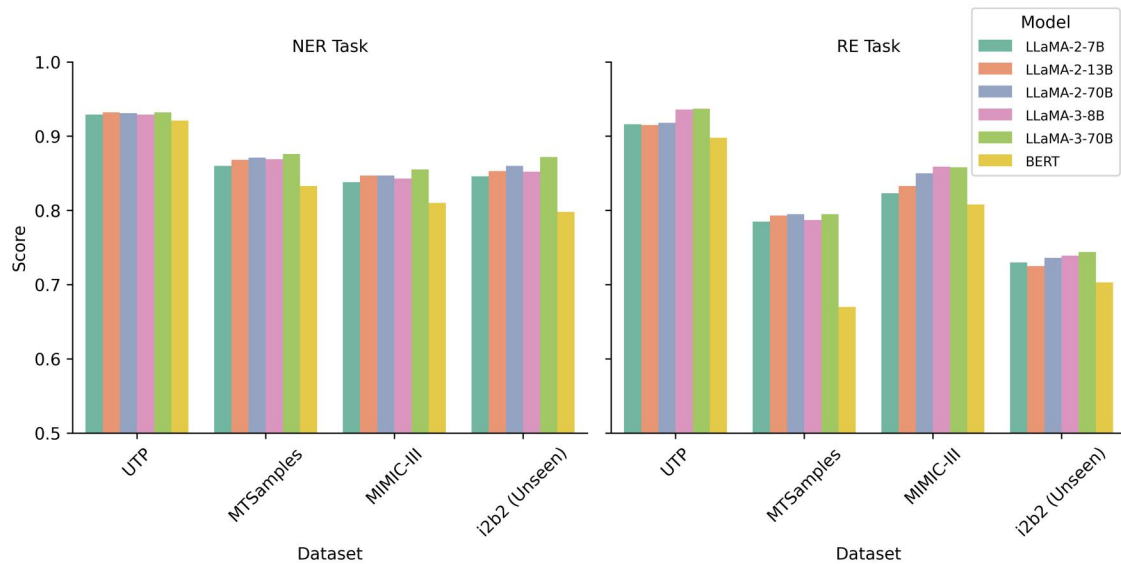


Figure 2. Exact F1 scores of LLaMA-2, LLaMA-3, and BERT on NER and RE across 3 datasets and cross-institution generalizability test results on unseen dataset (i2b2).

Table 2. Resource consumption comparison across different models using NVIDIA A100 80GB GPU.

	LLaMA2-7B	LLaMA2-13B	LLaMA2-70B	LLaMA3-8B	LLaMA3-70B	BERT
Memory usage (GB)						
Train	12.7	17.4	57.6	32.4	66.3	21.3
Test ^a	72.8	72.9	146.9	72.7	145.9	65.7
Total GPU hours (h)						
Train	41.7	71.2	320.2	38.6	273.5	96.8
Test ^a	0.5	0.9	2.2	0.4	2.2	0.08
GPU hours per epoch (h)						
Train	20.85	35.6	160.1	19.3	136.75	4.84
Energy consumption (kWh)						
Train	15.22	27.98	125.21	13.16	101.21	15.7
Test ^a	0.19	0.36	0.86	0.14	0.81	0.03
Carbon emissions (kg CO₂)						
Train	5.94	10.91	48.83	5.13	39.47	6.12
Test ^a	0.08	0.14	0.33	0.06	0.32	0.01

Training: 1413 documents; Test: 175 documents.

^a 70B models need 2 A100 80GB GPUs for testing.

Abbreviations: i2b2, Informatics for Integrating Biology and the Bedside; MIMIC-III, Medical Information Mart for Intensive Care; MTSamples, Transcribed Medical Transcription Sample Reports and Examples; UTP, UT Physicians.

Computational resources and throughput

We compared the computational resources and throughput of BERT, LLaMA-2, and LLaMA-3 using NVIDIA A100 80GB GPUs (Tables 2 and 3). Fine-tuning was conducted with 1 GPU for BERT and the LLaMA. For inference, BERT and the 7B, 8B, and 13B LLaMA models used 1 GPU, whereas the 70B models required 2 GPUs. Results show notable tradeoffs between performance and computational efficiency across model sizes.

Memory usage

Memory requirements increase substantially with model size. The 70B versions of LLaMA-2 and LLaMA-3 required 58-66 GB for training and 146 GB for inference, while BERT used only 21.3 and 65.7 GB, respectively. The 7B and 8B LLaMA models showed comparable memory efficiency to BERT, with training memory usage between 12.7 and 32.4 GB. Notably, during training, LLaMA models utilize low-rank adaptation (LoRA), which reduces memory requirements compared to

inference, where LoRA is not applied. This difference makes the memory needed for fine-tuning significantly lower than that required for inference. Additionally, these memory usages are heavily influenced by batch sizes; the current setup is optimized for best model performance.

Energy consumption and carbon emission

Energy consumption varied widely across models, with larger models requiring substantially more resources. LLaMA-70B models were the most energy-intensive during training, using 101-125 kWh and producing 39-49 kg of CO₂. Smaller LLaMA models and BERT were more efficient, using 13-28 kWh and 5-11 kg of CO₂. For inference, BERT was highly efficient at 0.03 kWh, while LLaMA models ranged from 0.14 kWh (LLaMA-3-8B) to 0.86 kWh (LLaMA-2-70B).

Training and inference throughput

Training and inference throughputs varied substantially across models and datasets as shown in Table 3. LLaMA-3-8B had

Table 3. Training and test times, and processing throughputs across models and datasets.

	LLaMA2-7B	LLaMA2-13B	LLaMA2-70B ^a	LLaMA3-8B	LLaMA3-70B ^a	BERT
Total training time (h)						
UTP	38.3	65.5	294.5	35.5	251.5	89.0
MTSamples	1.8	3.0	13.5	1.6	11.5	4.1
MIMIC-III	1.6	2.7	12.3	1.5	10.5	3.7
Training GPU hours per epoch (h)						
UTP	19.2	32.8	147.3	17.8	125.8	4.5
MTSamples	0.9	1.5	6.8	0.8	5.8	0.2
MIMIC-III	0.8	1.4	6.2	0.8	5.3	0.2
Training time per note (s/note)						
UTP	106.9	182.4	820.5	98.9	700.8	248.0
MTSamples	65.8	112.3	505.2	60.9	431.5	152.7
MIMIC-III	229.9	392.6	1,765.4	212.8	1507.9	533.7
Total inference time (s)						
UTP	725.7	1271.4	3009.2	582.1	2980.9	145.8
MTSamples	266.0	468.5	1157.9	232.4	1142.0	22.3
MIMIC-III	471.0	815.8	1998.6	406.9	2013.9	47.2
i2b2	398.8	695.4	1753.5	335.4	1786.2	81.6
Inference timer per note (s/note)						
UTP	14.5	25.4	60.2	11.6	59.6	2.9
MTSamples	5.3	9.4	23.2	4.6	22.8	0.4
MIMIC-III	18.8	32.6	79.9	16.3	80.6	1.9
i2b2	8.0	13.9	35.1	6.7	35.7	1.6

Average training note lengths: UTP: 978 tokens/note; MTSamples: 603 tokens/note; MIMIC-III: 2106 tokens/note. Average test note lengths: UTP: 1022 tokens/note; MTSamples: 655 tokens/note; MIMIC-III: 2086 tokens/note; i2b2: 822 tokens/note.

Abbreviations: i2b2, Informatics for Integrating Biology and the Bedside; MIMIC-III, Medical Information Mart for Intensive Care; MTSamples, Transcribed Medical Transcription Sample Reports and Examples; UTP, UT Physicians.

^a 70B models need 2 A100 80GB GPUs for inference.

the fastest training time at 38.6 GPU hours, while BERT required 96.8 GPU hours. Bidirectional encoder representations from transformers' longer training time can be attributed to its need for 20 training epochs, compared to only 2 epochs for LLaMA models. This difference is reflected in GPU hours per epoch: BERT stands out with a particularly low requirement of 4.84 GPU hours per epoch, whereas LLaMA2-7B and LLaMA3-8B require 20.85 and 19.3, respectively. However, LLaMA-70B models took much longer, requiring 273.5-320.2 GPU hours. Dataset characteristics such as note length impacted the training time; for example, MIMIC-III (2106 tokens) took significantly more time per note than MTSamples (603 tokens). Bidirectional encoder representations from transformers outperformed all LLaMA models in inference, processing notes 5-28 times faster. Among LLaMA models, LLaMA-3-8B was quickest, while the 70B models were slowest. These findings highlight a tradeoff: larger LLaMA models offer higher accuracy but are slower, while BERT is more efficient for time-sensitive applications.

Error analysis

Automatic and manual error analyses on LLaMA and BERT outputs across 4 datasets revealed a modest pattern: LLaMA-70B models tended to produce fewer false negatives (FNs), suggested by improved recall. While it slightly increased false positives (FPs), implying a marginal drop in precision. For instance, on the UTP dataset, LLaMA-70B reduced FN counts by 9.1% compared to BERT-base, but FP counts increased by 5.4%. These differences, based on raw error counts, suggest a tradeoff between recall and precision with model scaling.

Bidirectional encoder representations from transformers, while showing higher FN and FP rates, has fewer boundary errors, suggesting better accuracy in detecting text boundaries. Type errors were consistently low for all models,

meaning entity types are effectively classified. Manual review of 40 randomly selected sentences per dataset revealed that approximately 7.5% of gold-standard annotations were missing entities that the models correctly identified. This finding underscores the potential of LLMs to support annotation pipelines by highlighting instances of likely human omission.

Package implementation

To support real-world application, we created an open-source clinical IE tool, "Kiwi," featuring our LLaMA-based and BERT-based models. Kiwi, available at <https://kiwi.clinicalnlp.org/>, is Python-based and packaged as a Docker image for cross-platform compatibility. It includes a CPU version with BERT and a GPU version with both BERT and LLaMA models. Kiwi docker image supports batch processing, and an online demo with a GUI is also provided on the website for testing purposes (see Figure 3 for interface screenshots). Kiwi extracts medical entities and their relationships from user input, then converts the results into a JSON format readable by the frontend. For visualization, we use the BRAT annotation tool (<https://brat.nlplab.org/index.html>), which renders the extracted entities and relations as highlighted spans and labeled links for intuitive understanding. Kiwi's source code is released under the MIT License and hosted at <https://github.com/BIDS-Xu-Lab/Kiwi-LLaMA>. Because the underlying LLaMA weights are governed by Meta's noncommercial academic license, we distribute the fine-tuned checkpoints on request for research use via <https://kiwi.clinicalnlp.org/download/>.

Discussion

This study presents one of the first comprehensive evaluations of instruction-tuned LLMs for clinical IE tasks using diverse institutional datasets, comparing them with previous

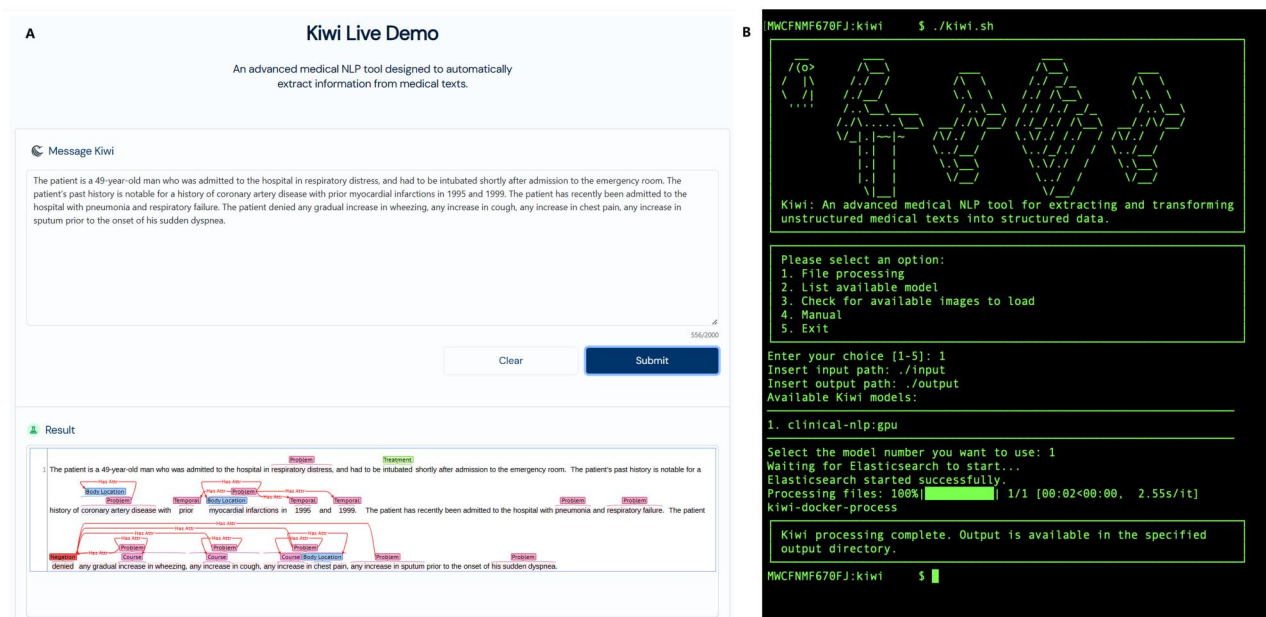


Figure 3. Screenshots of Kiwi's Web Interface (A) and Command Line Interface (B).

SOTA BERT models. We address several limitations in existing LLM-based clinical IE research: previous studies often focused on datasets with limited entity types and relation types,^{29,33} used small corpora for evaluation,⁴⁸ and primarily tested on similar types of clinical texts.^{29,48} By developing a broad, multi-institutional corpus including a wide range of medical contexts, entity types, and relation types, we offer a more comprehensive and generalized assessment of LLMs' capabilities in real-world clinical IE scenarios.

Generative AI, especially LLMs, have gained immense popularity and have opened up vast potential across various industries. In health care, there is growing interest in utilizing LLMs to develop and implement IE systems that support both clinical research and practice. However, our study highlights that several challenges still impede the broad adoption of LLMs for clinical IE tasks. Despite their impressive performance, issues such as high computational costs and slow processing throughputs remain significant barriers. In the following sections, we discuss how to choose between LLMs and traditional deep learning approaches for clinical IE applications, factoring in performance, throughput requirements, and the availability of computational resources.

Performance and generalizability

The consistent outperformance of LLaMA models, especially on the low-resource datasets (MTSamples and MIMIC-III) and on unseen dataset (i2b2), underscores their potential to address the critical challenge of variability in clinical documentation.⁴⁹ This variability, stemming from factors such as local practices, specializations, and patient demographics, has hindered the widespread adoption of NLP solutions in health care. The ability of instruction-tuned LLaMA models to generalize effectively across these variations suggests a promising path toward more universally applicable clinical NLP tools.

However, it is important to note that while the performance gains are significant, they are not uniform across all

tasks and datasets. When abundant annotated data are available at a health-care system (eg, UTP), the performance improvements of LLaMA are modest when compared with the BERT model (0.932 vs 0.921 in F1 for NER and 0.937 vs 0.898 in F1 for RE), suggesting that deep learning models like BERT remain competitive in this scenario. Considering the low cost and high throughput of BERT models, we would recommend BERT models over LLMs when substantial annotated training data is already available for the IE task. However, when annotated training data are limited to a health-care system (eg, the case of i2b2), LLMs could be considered first, as they show better generalizability, indicating less annotation is needed to build the IE system. This highlights the need for careful consideration of the specific use case and available data when selecting a model for deployment.

Throughput

Our study highlights a significant difference in processing throughput between LLM-based and BERT-based systems, which is crucial for real-world clinical IE applications. In settings where real-time processing is required, such as during patient care, slow NLP systems are impractical as users cannot wait for delayed results. Similarly, in research environments handling millions of clinical notes, the lower processing throughput of LLMs makes them less feasible, even if they offer slightly better performance. Ultimately, the choice between LLM- and BERT-based systems must consider both performance and throughput, as the latter is often task-specific and essential in many clinical and research scenarios.

Computing resources

The comparative analysis of computational requirements reveals important considerations for both inference and training. For inference deployment scenarios: (1) in high-resource settings with multiple NVIDIA A100 GPUs or better, LLaMA-3-70B offers the best performance; (2) in balanced settings with a single A100 GPU or equivalent, LLaMA-3-8B

	High-resource setting (Multiple NVIDIA A100 80G GPUs or better)	Balanced setting (Single NVIDIA A100 80G GPU or equivalent)	Resource-constrained setting (Entry-level GPU or No GPU)	Overall recommendations:
Data Availability	LLaMA-3-8B or BERT depending on the availability of annotated data (For the tasks with sufficient annotated data (e.g., thousands of notes), BERT shows a similar performance to LLaMA. But with limited data (e.g., 25 notes), LLaMA models show a more decent performance.)		BERT is the only recommended model	1. LLaMA-3-8B is generally recommended if computational resource permits, as it provides the balance between performance and throughput.
Generalizability Requirement	LLaMA-3-70B or 8B is recommended (both generalize very well on unseen dataset)	LLaMA-3-8B is recommended (considering trade-off between performance and process time)		2. BERT is recommended for time-sensitive tasks, or in resource-constrained setting.
Speed Requirement	BERT models is recommended (or depends on # of notes and note length, estimate time needed for each according to Table 4)			3. LLaMA-3-70B model is recommended for cross-institutional tasks in high-resource setting.
				4. LLaMA-2 models are generally not recommended.

Figure 4. Recommendations for model choice for different tasks in different resource settings.

provides an attractive middle ground, achieving significant performance improvements; (3) for resource-constrained settings, BERT remains viable, requiring less resources while maintaining competitive performance on in-domain datasets. For training considerations, institutions have several options: (1) using parameter-efficient tuning on larger LLaMA models (70B) demands significant resources but achieves the best performance; (2) training smaller LLaMA models (7B/8B/13B) offers a balanced approach with competitive results; (3) traditional fine-tuning of BERT provides the most resource-efficient though with lower performance on cross-institutional data. For institutions with extensive computational resources, pretraining domain-specific models like Me-LLaMA represents another option, though this requires more, making it practical only for large-scale research environment.⁵⁰ These tradeoffs underscore the importance of carefully matching computational capabilities with specific use case requirements.

Based on these findings, we made some recommendations for model choice for different clinical IE tasks in different resource settings, summarized in Figure 4.

While our study focuses on fine-tuning, the pretraining cost of foundation models is also relevant. For instance, Meta reported that pretraining LLaMA-3-405B took 54 days on 16 000 H100 GPUs, representing a massive upfront resource investment. In contrast, BERT-base was pretrained in roughly 4 days using 16 Cloud TPUs. Although LLaMA-2/3-7B models are smaller, the relative cost gap remains large. This highlights the importance of amortizing pretraining costs across many downstream tasks. Unfortunately, exact figures for most models remain proprietary, limiting full transparency.

Another significant contribution of this study is to release the Kiwi package. To the best of our knowledge, Kiwi is the one of the first open-source clinical IE systems based on LLMs. Kiwi includes both our instruction-tuned LLaMA-based and BERT-based models, available in both CPU and GPU versions to accommodate the diverse computational resources of different health-care institutions. By offering different versions of models, Kiwi enables institutions to select the most appropriate model based on their specific needs, available resources, and use cases according to our suggestions. As a pioneering LLM-based clinical IE system, Kiwi represents a significant milestone in the application of

generative AI to clinical IE. By making our system publicly available, we aim to accelerate research in LLM-based clinical IE by providing researchers with a ready-to-use framework. This will enable the medical NLP community to use our models directly, customize models upon our work, and develop more advanced clinical IE solutions.

While our study offers valuable insights into the potential of LLMs for clinical IE, future research should explore knowledge distillation techniques to create smaller, more efficient models better suited to resource-constrained health-care settings.^{51,52} Continual training on institution-specific data may also enhance model performance by adapting to local documentation styles and terminology.

Several limitations merit consideration. First, our IE approach operates at the sentence level, which may not fully utilize LLMs' ability to process longer sequences. Future work could explore section- or document-level modeling to capture more complex intersentence relationships. Second, we used LLaMA-2/3 models not pretrained on medical text; instruction-tuning domain-specific LLMs (eg, Me-LLaMA⁵⁰) may yield improved performance. Third, moderate interannotator agreement (75.4%-82.5%) may have introduced noise, particularly in relation extraction, though LLaMA models remained robust. Lastly, our cross-institutional dataset, while diverse, is geographically limited to the Boston area, potentially affecting generalizability. Although i2b2 notes and MIMIC-III are both under data use agreements, we cannot rule out their potential inclusion in LLM pretraining via unofficial sources. MTSamples.com (the website hosting the MTSamples dataset) is included in the Common Crawl listing, which was heavily weighted within LLM training sets. We therefore acknowledge the possible limitation to our evaluation in that all non-UTP datasets were likely included in developing the base LLM models.

Conclusion

In summary, our evaluation demonstrates that instruction-tuned LLaMA models hold significant promise for clinical IE tasks, particularly in challenging low-resource and cross-institution scenarios. The empirical results indicate that LLaMA models can achieve notable improvements in both

NER and RE compared to traditional BERT-based systems. However, these gains come with increased computational costs and slower processing speeds—a tradeoff that health-care institutions must consider when integrating such technologies into real-world applications.

Model selection should balance performance, generalizability, throughput, and resource constraints. For environments with abundant annotated data and limited computational resources, BERT-based approaches remain a competitive option. Conversely, in settings where data are scarce or generalizability across diverse clinical narratives is crucial, LLaMA models offer a compelling alternative despite their higher resource consumption.

To bridge the gap between research and practice, we have released the Kiwi package, which provides both LLaMA-based and BERT-based clinical IE solutions. By making Kiwi publicly accessible, we aim to facilitate further innovation and adoption of advanced NLP techniques in the clinical domain, ultimately contributing to more efficient and accurate health-care data processing. Future directions include model distillation and domain-specific pretraining to improve efficiency and applicability in diverse settings.

Author contributions

Y.H. and H.X. contributed to the conception and design of the study. Y.H. and X.Z. performed the experiments and analyzed the data. Y.Z. led the data annotation. Y.H., Q.C., X.Z., C.S., and H.X. contributed to the interpretation of the results. Y.H., V.J., and J.H. contributed to the development of the Kiwi package. Y.H., J.H., V.J., V.K., and R.L.W. contributed to the interface development. Y.H., Q.C., X.Z., X.J., K.R., V.K., C.S., and H.X. drafted the manuscript and revised it. All authors approved the final version of the manuscript and agreed to be accountable for all aspects of the work.

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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Conflicts of interest

The authors have no competing interests to declare.

Data availability

The MTSamples dataset can be directly accessed on our GitHub repository. For the MIMIC-III and 2010 i2b2 datasets, we are providing entity offsets and texts without the original text files on our GitHub repository. Researchers can obtain the original text files from their official websites. The Kiwi codebase and annotation guidelines are publicly available under the Apache 2.0 License at our GitHub repository: <https://github.com/BIDS-Xu-Lab/Kiwi-LLaMA>. The Kiwi tool, including the fine-tuned models, is available upon

request via the official Kiwi website: <https://kiwi.clinicalnlp.org/>. For the base language model, we use Meta's Llama 3 model, which is released under the Meta Llama 3 Community License: <https://www.llama.com/llama3/license/>.

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